ELEC70057
ELEC60005

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2022

BIOMEDICAL ELECTRONICS

Tuesday, 14 December 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer 4 out of 5 questions

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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1. This question relates to the design of an instrument for measuring bio-potentials.

   a) Sketch a typical ECG waveform labelling each segment. Explain briefly what each segment of the ECG waveform represents during a single heartbeat.

   [4]

   b) Table 1.1 lists five commonly observed bio-potentials and their amplitude and frequency characteristics. A doctor requires an instrumentation system to record the ECG from the heart.

   Design a complete instrumentation system for recoding ECG from the chest according to the following specifications:

   Instrumentation Amplifier gain: 200
   Low-Pass Filter: 3rd Order
   High-Pass Filter: 2nd Order
   Supply Range: -3V to 3V

   State all assumptions and values of resistors and capacitors used. You may assume that the instrumentation system utilizes the whole dynamic range available and 10nF capacitors are available for the filters.

   [8]

   c) Design a circuit to reduce the common mode voltage created by current capacitively coupled to the body from the power-lines and derive the equation showing how this improvement has an effect.

   [4]

   d) After measurement of the ECG you notice that the instrumentation amplifier keeps saturating due to a large DC offset voltage between the input electrodes. The impedance of the input electrodes is 1MΩ. Sketch a modified instrumentation amplifier able to solve this problem, stating suitable component values where necessary.

   [4]

Table 1.1

Signal	**Frequency range (Hz)**	**Amplitude range (mV)**
ECG	0.1 – 300	0.05 – 3
EEG	0.5 – 40	0.001 – 1
EMG	20 – 2000	0.001 – 100
Neural Spikes	300-5000	0.001– 0.5
Local Field Potentials	10-200	0.001 – 5

2. This question relates to the design of a deep brain stimulator (DBS):

a) Discuss the two main requirements of a stimulation circuit. [4]

b) Why do DBS electrodes typically have four electrode contacts? [3]

A typical DBS system needs to generate a symmetric biphasic stimulus in range 0.5-2.5mA, with pulse width in range 60-250μs, at a repetition rate of 125-500Hz.

c) Estimate the device lifetime for a DBS implant with 6.3Ah primary cell battery. If you were to develop a rechargeable version of this device, what rechargeable battery capacity would you select? Explain your rationale. [7]

d) Sketch a possible design (system architecture, i.e. block diagram) for this DBS device. [6]

3. This question relates to the design of a novel integrated sensor for chemical measurement.

a) Electrochemical measurement has as a basis a redox reaction. Explain what is meant by a redox reaction using reaction equations where necessary.

[4]

b) Figure 3.1 shows a new pixel architecture to be used as part of an array in an integrated CMOS based diagnostic device. The total pixel size is 150µm x150µm with a total sensing area of 100µm x 100µm which is capable of sensing both glucose and pH. The working electrode is coated with glucose oxidase to give a peak current to glucose at V_{Cell}=0.7V according to the following reaction:

$$\text{glucose} + O_2 \xrightarrow{\text{glusoce oxidase}} \text{gluconic acid} + H_2O_2 \quad (1)$$

$$H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$$

i) The sensor shall be used to measure a physiological glucose range of 0-20mM. The peak redox current for a scan rate of 1V/s is given by the following equation:

$$i_p = 2.69 \times 10^5 n^{3/2} A D^{\frac{1}{2}} C$$

where n is the total number of electrons given in equation 1, the diffusion coefficient D=4.38X10^{-8} m^2/s and A is the total area of the working electrode, WE, in Figure 3.1. Calculate the maximum range of the WE current, using the dimensions shown in Figure 3.1, to capture the physiological range of glucose.

[2]

ii) The amperometric sensor in the pixel shall be driven by a potentiostat and transimpedance amplifier which uses an integration capacitor, C_E to convert the sensed current to a voltage. The circuit operates off a single supply with V_{DD}=3.3V and V_{SS}=0V. Sketch the schematic of a suitable measurement system used to interface to the amperometric sensor and label the voltages of all electrodes.

[6]

iii) The transimpedance amplifier has a poly-silicon integration capacitor determined by the remainder of the area in the pixel shown in Figure 3.1. A capacitance per unit area of poly-silicon is 4x10^{-4} pF/µm^2. Calculate a suitable clock frequency that maximizes the dynamic range of the pixel for glucose measurement.

[4]

iv) Sketch a suitable schematic to interface to the pH sensor and explain its operation.

[4]

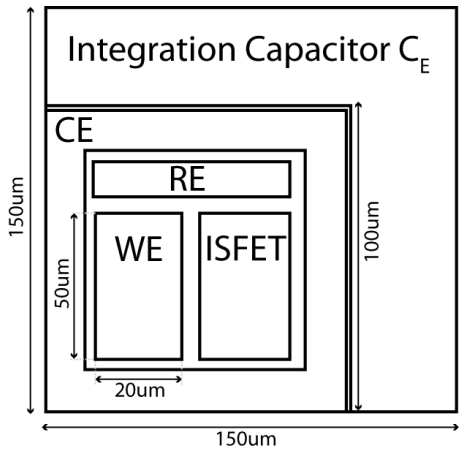

Integration Capacitor C_E
CE
RE
WE
ISFET
150um
50um
100um
20um
150um

Figure 3.1

4. This question relates to a cortical prosthesis for vision:

    a. Compare the relative merits/drawbacks between a retinal and cortical visual prosthesis. [4]

    b. Given the prosthesis will utilise penetrating electrode array (e.g. Utah), how will this influence the stimulator circuit specifications? Provide estimated values for key related specification. [6]

    c. Given the electrode material is platinum iridium (charge injection capacity ~200μC.cm^{-2} per phase), estimate the maximum stimulation power consumption for a 1000-channel cortical prosthesis. State any assumptions made and describe your thought process. [8]

    d. Describe which of the assumptions made in your answer above to (c) is likely to pose the major limitation to a visual prosthesis. [2]

5. Sequencing the human genome requires the identification of 3.2 billion bases of DNA which can be implemented using the process of pyrosequencing.

    a) Explain what DNA is and describe one of its physical properties that allow detection of single bases for DNA sequencing.

    [4]

    b) Explain the process of pyrosequencing, including a diagram showing the sequence of DNA bases with the signals generated for detection.

    [4]

    c) Name one sensor which is suitable for detecting DNA bases, drawing the structure of the device and listing the equations determining its operation.

    [4]

    d) Draw a block level circuit schematic to show how the sensor named in part c can be connected when scaled to millions of sensors.

    [5]

    e) Explain why using this sensor has significantly reduced the cost of DNA sequencing.

    [3]